

Forecasting for optimization in road freight transport: A review

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ABSTRACT

In operations of road freight transport, demand forecasts are mostly used for downstream optimization, following a predict then optimize setting. While the literature provides numerous optimization models that rely on accurate forecasts to support the planning of road freight transport, dedicated work on forecasting is limited, hindering a holistic treatment of the problem. Moreover, there is a disconnect to advances in the broader forecasting literature, limiting the adoption of modeling innovations and methodological advances. These can harm the quality and validity of forecasts designed to support road freight transport. We link the relevant forecasting publications to different prominent optimization problems for road freight transportation, highlighting disconnects between forecasting and optimization models in the area. By contrasting these with the current discourse in the forecasting literature, we identify relevant modeling and methodological improvements, in model building, the choice of loss function, and evaluation. These are important to better link forecasts with optimization models for road freight transport. Furthermore, we propose a unified hierarchical framework for freight demand to support aligned decisions across planning levels, which is an important consideration for practice. Our review helps structure and steer academic research and provides opportunities to improve existing forecasting models; a closer integration with optimization; and ultimately improve decisions.

1. Introduction

Road freight transport planning problems involve uncertain demand and are often handled in a “predict then optimize” fashion. Road Freight Demand (RFD) forecasts are generated and then serve as input into optimization models that provide decisions. We define RFD as the demand to move freight from an origin to a destination via road transport from the viewpoint of a logistics enterprise. Forecasting and optimization models are becoming increasingly relevant in this domain, as enterprises put substantial effort into digitization, and automation or support of decisions, which has led to substantial cost savings in operations (Gorman et al., 2014; Dohrmann et al., 2022). At the same time road transport remains along the most important modes of transportation, for example, accounting for 77% of the overall transported freight volume in Germany (Statistisches Bundesamt, 2022, p. 7), and amounting to more than 13.2 billion tonnes and 1 857 billion tonne-kilometers for the whole European Union (Eurostat, 2024). Market reports claim that the global road freight transport market will increase from USD 559.83 billion in 2023 to USD 802.55 billion in 2031 (Markets & Data, 2024). At the same time, in 2020, around one-fifth of the vehicle-kilometers in the EU were empty runs (Eurostat, 2021), pointing towards large inefficiencies in the operations of road freight carriers.

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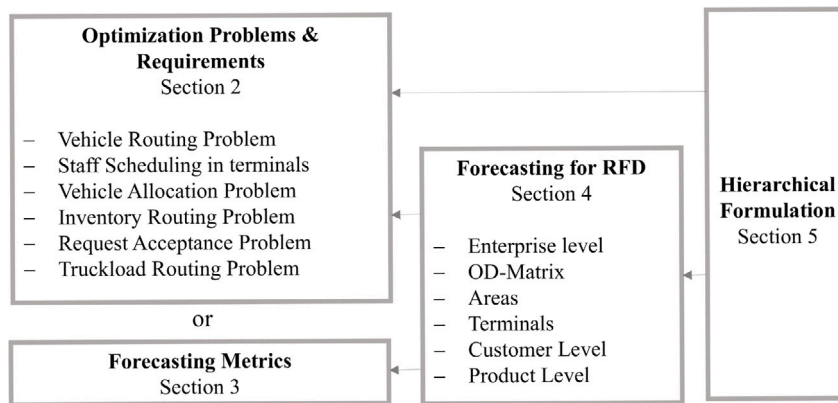


Fig. 1. The sections of the paper, mapped on the predict then optimize framework steps.

While the optimization literature provides numerous models to support the planning for road freight enterprises and thus improves its efficiency, this domain has only received limited attention by the forecasting community. However, it poses specific requirements for forecasting, which arise out of domain-specific planning processes, such as freight that fills only containers partially, last mile logistics, and a network design that is more flexible compared to ship or rail (Gorman et al., 2014). As there are several challenges on the interface of forecasting and optimization (see for examples, Kourentzes et al., 2020; Elmachtoub and Grigas, 2022) and the latter ultimately gives the decisions, we argue that forecasting should be guided by its use in optimization. Accordingly, we motivate the discussion by the supported optimization problems, enabling us to highlight the requirements placed on the forecast. In doing so, we discuss unique opportunities for RFD forecasting stemming from research into hierarchical forecasting, and show how it can be implemented in RFD forecasting in Section 5.

Fig. 1 visualizes the predict then optimize set up, where first forecasts are generated and then fed into an optimization model that delivers a decision. First, in Section 2 we link the RFD forecasting literature to the optimization literature in this domain, which is important in identifying the appropriate objectives for forecasting models in RFD. We focus on the forecasting objective, data granularity, forecast horizon (Section 2.1). The treatment of uncertainty has important implications on the optimization formulation and set requirements for forecasting. In Section 2.2 we discuss alternatives and their linkage with forecasting. Most surveyed studies do not evaluate the forecast in connection with an optimization model, relying on reporting improved forecasting performance. In Section 3, we discuss the appropriate metrics for forecasting performance to match the supported optimization models and the requirements of RFD decisions. In doing so, we take a critical view of past forecasting work in RFD, demonstrating a mismatch between requirements and proposed solutions. In Section 4 we build on these to take a critical view of the forecasting literature in RFD. First, we structure it depending on the sub-application area, summarize key findings, and evaluate their validity by considering how well it aligns with the requirements of optimization and the use of appropriate metrics. Lastly, in Section 5, we develop a unifying hierarchical framework for RFD. The surveyed forecasting and optimization models rely on different demand aggregations and are performed in isolation, when supporting the various RFD decisions. This can result in mismatching plans, even though it is crucial for organizations that these are aligned. Drawing on the current state-of-the-art in the forecasting literature, we propose hierarchical forecasting as an approach to support joint and aligned planning, by ensuring coherent forecasts that consider multiple data granularities, objectives, and horizons. This has the potential to address limitations identified in the literature in the preceding sections, opening new research directions for both forecasting and optimization models.

2. Linking RFD forecasting to RFD optimization

The optimization problems for road freight transport enterprises arise out of the business process that enables the transportation of freight. In general, these business processes can be classified into: (i) Full Truck Load Carriers (FTL Carriers), (ii) Less Than Truckload Carriers (LTL Carriers), and (iii) Parcel Carriers (Roy and Crainic, 1992). For FTL Carriers a shipment requires a whole trailer. Hence, a truck is loaded at the shipper and then driven directly to the receiver, where it is unloaded. From the receiver, it is either driven empty to the next shipper, where new freight is loaded, or it is returned empty to the original shipper. Fig. 2(a) illustrates this process.

LTL carriers handle shipments that typically weigh between 50 kg and 4.500 kg, which means that they do not fill a whole truck and, therefore shipments are consolidated in terminals (or *cross docks*) to utilize the full capacity of trailers. LTL-shippers build up a network of origin, transfer, and destination terminals. In a tour, a truck collects the shipments of multiple shippers and delivers them to the origin terminal. In the origin terminal all the shipments that go towards the same transfer terminal are consolidated onto another truck. This truck is then driven fully loaded to the transfer terminal. Here, freight that goes to the same destination terminal is consolidated in the next truck. From the destination terminal, then finally shipments are delivered in a tour that covers multiple customers. Fig. 2(b) illustrates this process. The transportation process is similar for parcel services, however, shipments typically weigh below 32 kg.

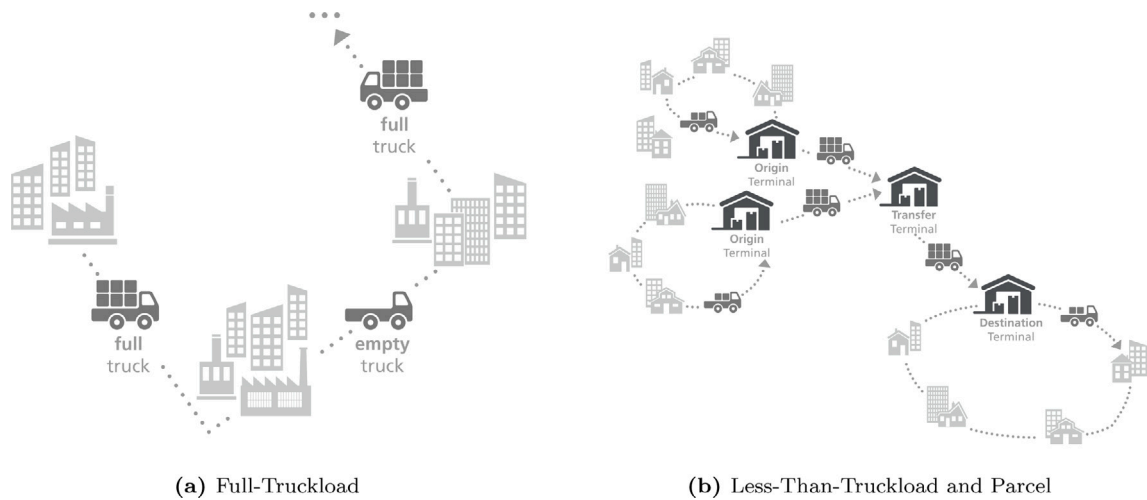


Fig. 2. Schematic process of the three areas of application.

For each application, operations need to be planned on various levels, and correspond to specific optimization problems. In the following Section 2.1, we give an overview of the prominent optimization problems in road freight transportation and link the respective forecasting literature to the described problems. We consider especially the treatment of uncertainty in forecasting and optimization in Section 2.2.

2.1. Attributing forecasting publications to optimization problems

Most of the forecasting literature does not explicitly relate the proposed forecasting models to the optimization models served. Acknowledging that, we assess to what extent the forecasting literature addresses the forecasting problems that are posed by optimization, to provide relevant inputs to predict then optimize solutions. To identify relevant RFD forecasting literature, we sourced publications from EBSCO, Web of Science, and SCOPUS with the search term: *(forecast* OR regression OR predict* OR plan) AND (freight OR logistic* OR truck* OR transport* OR carrier OR shipper OR shipment) AND demand*. We limited our search to papers where the key-values appeared in the title, abstract, or key words. Further, we restricted our search to areas that relate to freight demand forecasting (e.g., we excluded papers assigned to chemistry). We only considered publications in English or German. Our search resulted in 15 759 papers. These were independently screened by two researchers for relevance, first in their titles, and the remaining ones in the abstracts, and then again the remaining ones on the full text. We did not restrict to peer-reviewed journals due to the very limited available literature on RFD forecasting. Instead, we provide along with the findings in the literature an assessment of their validity. As relevant, we defined papers that deal with freight demand forecasting from the viewpoint of a motor carrier, as is commonly understood in operations research. We did not include papers that deal with freight demand forecasting from a policy-maker point of view. This resulted in 21 publications. Using forward and backward search, we found 9 more relevant publications (of which we were not able to retrieve Zhou et al., 2005). Given this relatively small available literature in the domain, we reflect upon the broader forecasting literature throughout our review.

In Table 1 we link forecast publications to prominent optimization problems for motor carriers. These can be split in strategic, tactical and operational decision making. Strategically, carriers design the physical infrastructure of their network, thereby deciding where to locate facilities, what type they should be, and what lines to serve (Crainic and Laporte, 1997). For example, Dachser, one of the biggest German carriers, operates terminals all across the nation, whereas smaller carriers organize themselves in cargo networks. This requires a long-term RFD forecast for the next years on a spatially aggregate level. Tactically and operationally, carriers determine how to operate in their network (Crainic and Laporte, 1997). Tactical planning requires forecasts for the next season, for example, to determine the transportation schedules between terminals or train cross-dock staff. Operational planning requires short-term forecasts, on a granular level. For example, the workforce in cross docks may be adjusted based on the daily fluctuation of workload.

In the following, we describe briefly some central operational and tactical optimization problems in road freight transport and point out the mismatches to the attributed literature. Where we see even a vague connection on the basis of the granularity of the forecast, we attribute the paper to the respective optimization problem. This naturally can put a contribution in a category that is beyond the stated objectives of a paper, and accordingly the following gaps we point out should not be seen as critique to the authors. Of the 29 examined publications we could link 17 to optimization problems. However, mismatches arise on the forecast horizon, unit, and granularity. The vehicle routing problem, staff scheduling in terminals, and the dynamic vehicle allocation problem are barely addressed by the forecasting literature, and no publication evaluates forecasts for the demand variable exactly as it appears in any of these. We describe them nevertheless due to their major role in the planning processes of road transport.

Table 1

Linking decisions, source of uncertainty, optimization and forecasting.

Optimization (% of papers)	Decision	Source of Uncertainty	Required forecast	Forecast publications
(Stochastic) vehicle routing problem (7%)	Assignment of trucks to routes in short distance traffic	Volume of freight at the customer nodes	Customer level	Blumenfeld and Beckmann (1988), Tabib et al. (2022)
(Stochastic) Truckload Routing Problem (3%)	Assignment of trucks to routes for FTL	Uncertain short term RFD on FTL relations	FTL OD-Matrix	Miori (2009)
(Stochastic) Staff scheduling in terminals (7%)	Number of employees needed the next week	Workload at the terminal	Terminal level	Grzelak et al. (2019), Najafabadi (2022)
(Stochastic) Service Network Design Problem (3%)	Service schedule between terminals	Uncertain long term RFD on relations	OD-Matrix between terminals (mid-term horizon)	Al Hajj Hassan et al. (2020)
(Stochastic) Vehicle Allocation Problem (17%)	Short term assignment of vehicles to relations (FTL or LTL)	Uncertain short term RFD on relations	OD-Matrix between terminals (short-term horizon)	Godfrey and Powell (2000), Miori (2009), Mrówczyńska (2013), Karis (2019), Al Hajj Hassan et al. (2020)
(Stochastic) Inventory Routing Problem (10%)	Deliveries to distribution centers by truck	Uncertain demand for stock keeping units at the distribution centers	Product-Level	Chen and Lin (2009), Assemi et al. (2019), Deutsch et al. (1994)
Request Acceptance Problem (17%)	Acceptance of freight orders	Uncertain future requests and their profitability	OD-Matrix or delivery areas	Cleophas and Ehmke (2014), Lin et al. (2018), Garrido and Mahmassani (1998), Garrido and Mahmassani (2000), van der Meulen (2019)
Unassigned (41%)	–	–	–	Fite et al. (2002), Lu et al. (2007), Liu and Lu (2008), Zhou (2017), Mitkow and Świderski (2019), Ren et al. (2020), Alqatawna et al. (2023), Gao et al. (2010), Kmiecik (2021), Yuan et al. (2018)

Vehicle routing is needed for the planning of pick-up from the origin and delivery to the destination at the regional level around terminals (Crainic and Laporte, 1997). It is used to decide which truck serves which customers in what order (Sharma et al., 2018). Over the past decades a number of variations of the problem have been introduced (see, Toth and Vigo, 2014; Ritzinger et al., 2016; Oyola et al., 2018; Soeffker et al., 2022), encompassing static, dynamic, stochastic, and dynamic and stochastic vehicle routing problems (Schorpp, 2011; Ritzinger et al., 2016). In dynamic vehicle routing problems, the routing decisions are continuously updated. When the uncertain parameters are observed, e.g., gradually the actual freight volume of the customers along the routes is revealed, replanning is done accordingly (Soeffker et al., 2022). Stochastic vehicle routing problems account for the uncertainty in one or more parameters when planning routes, e.g., travel times or demand volume, with the latter being our focus here (Ritzinger et al., 2016). Consequently, the cumulative demand along a constructed route is stochastic and there is a need to balance recourse costs, when a truck is full before the end of the route, and opportunity costs, when a truck could have handled another stop on one route.

Blumenfeld and Beckmann (1988) forecast the number of stops per tour based on assumptions of demand distributions per customer. However, no forecast per customer is made which would be required by this optimization problem and in reality the disaggregate per customer series are expected to be substantially different than the aggregate ones. Tabib et al. (2022) make a forecast per customer to identify specific customers where the truck utilization at that point in the route is low and where origin points (retailers) are close. The optimization and prediction model are implicitly intertwined, since the routing in the vehicle routing problem is decided on the expected freight volume at each customer. Moreover they do not provide predictive distributions or quantiles as required, for example, by the probabilistic formulations of Gendreau et al. (2014). Predictive distributions provide an uncertainty estimate in the form of a distribution along with the point mean forecast (for further explanation, see Saoud et al., 2022).

The *Truckload routing problem* solves the routing of full trucks from origins to destinations (Miori, 2009). The demand can be regarded as stochastic. In the formulation of Miori (2009) the demand appears as hard constraint (see Section 2.2) for certain time windows. Miori (2009) evaluates assumptions about distributions, but does not evaluate forecast accuracy or impact on decision making (see Section 3).

Staff scheduling in terminals: Terminals in LTL transportation are used to consolidate shipments for line-haul movement (long distance transport; Roy and Crainic, 1992). In origin terminals the trucks for short distance transport are unloaded, the shipments are sorted and loaded to line haul trailers (vice versa for destination terminals; Roy and Crainic, 1992). In practice, these operations are

still mostly performed manually, making staff scheduling one of the most relevant planning problems for terminal handling (Ladier et al., 2014).

Practitioners need to plan the allocation of the workforce to meet the incurring workload, which poses a challenge, especially as the workload varies a lot over time (weekly, hourly, and daily; Ladier, 2014; Grzelak et al., 2019). Logistics service providers react to those variations by hiring temporary workers or using overtime compensation (Ladier, 2014). The actual workload can thereby be unknown at the time of planning and needs to be derived from a RFD forecast, since it is based on the freight volume that needs to be loaded or unloaded. Grzelak et al. (2019) forecast the demand for pallet space in terminals but this cannot be directly translated into workload for loading and unloading to support this optimization problem.

(Stochastic) service network design problem: The transportation between terminals is mostly determined by a load plan, which results out of the service network design problem (Lium et al., 2009). This schedule defines a planning period (e.g. a week) and is then fixed for the planning horizon (e.g. the next season, Baubaid, 2020). Crucial in the planning process at this stage is the anticipated (and hence forecasted) freight demand between terminals (Lium et al., 2009). This freight demand between terminals is not known at the time of planning and its deviation from the forecasted value can heavily influence the quality of the found solution (Lium et al., 2009). Al Hajj Hassan et al. (2020) forecasts RFD in the Origin–Destination–Matrix (OD-Matrix) formed by the lanes between terminals up to two months ahead. Nonetheless, service network design needs a forecast for the entire operating season (Baubaid, 2020).

The (stochastic) dynamic vehicle allocation problem: For a shorter planning horizon the load plan is dynamically adjusted based on the updated demand forecasts (Vasco and Morabito, 2016; Karis, 2019, p. 16). This is done in the dynamic vehicle allocation problem. In the LTL-case the number of empty and full trucks that should move between terminals is calculated, as well as the trucks that should be held at a terminal for a time period (Vasco and Morabito, 2016). In comparison to vehicle routing and staff scheduling, there is only limited literature for this problem, however, it is central for road freight transport enterprises.

Godfrey and Powell (2000) and Garrido and Mahmassani (1998, 2000) do not evaluate on lane level, though they frame their method in the context of forecasting the OD-Matrix. The forecast evaluations of Al Hajj Hassan et al. (2020), Karis (2019), and Mrówczyńska (2013) match the required horizon and granularity, however, they evaluate only on a small sub-sample of the OD-Matrix, whereas a forecast for the whole OD-Matrix would be required by the formulations of Powell (1986) and Vasco and Morabito (2016). Nonetheless, these are easily addressable issues.

Inventory routing is an extension to the vehicle routing problem, where the inventory at the customers is known. This setting arises in vendor managed inventories (Coelho et al., 2014). It differs slightly from the described processes of road freight transport carriers, since those usually do not control the inventory of their customers. The aim of the decision maker is to manage the deliveries to the customers so that the inventory holding cost and the transportation cost are minimized. In that manner, they decide when to serve which customer, how much to deliver to those customers, and how to combine customers into routes (Coelho et al., 2014). The attributed literature matches the requirements.

The *request acceptance problem* formulates the issue of dispatchers whether they accept or refuse a request for freight transportation with a certain quota (Schönberger and Kopfer, 2012). If not all requests are known at the decision time, the acceptance problem is dynamic. Cleophas and Ehmke (2014) assumes a B2C-parcel market and evaluates the impact of forecast accuracy on capacity planning in a revenue management setting. Both mentioned contributions base their findings on simulated demand data. The evaluation of Lin et al. (2018) is unclear in its details.

Unassigned forecasting papers: We could not assign 41% of the papers to any of the prominent optimization problems in transportation. Most mention a planning problem but do not refer to an identified decision in the optimization literature (Fite et al., 2002; Lu et al., 2007; Liu and Lu, 2008; Zhou, 2017; Mitkow and Świdorski, 2019; Ren et al., 2020; Alqatawna et al., 2023). Three papers do not mention any served planning problem (Gao et al., 2010; Kmiecik, 2021; Yuan et al., 2018).

2.2. Treatment of uncertainty in forecasting and optimization

In all the above optimization problems, RFD appears either as a hard constraint, where a restriction ensures that enough resources are allocated or reserved, or as a soft constraint, where over- and/or under-allocation of resources is penalized (Ladier et al., 2014; Toth and Vigo, 2014; Vasco and Morabito, 2016). These are in general the dominant modeling paradigms (e.g., in staff scheduling, van den Bergh et al., 2013). Mixtures between the approaches are possible, for example, the over-staffing of employees might be penalized, while under-staffing may be forbidden (Defraeye and van Nieuwenhuyse, 2016). Both approaches can be formulated in a deterministic manner, where demand uncertainty is ignored, or a probabilistic manner that includes some estimate of the demand uncertainty.

Table 2 summarizes the two approaches and their probabilistic adaptations. Chance constraints typically restrict upon a quantile of the uncertain variable. This can be evaluated via the pinball loss (Gneiting and Raftery, 2007; Kolassa, 2023). In stochastic programming, the relation between forecast errors, including wrong uncertainty estimates, and cost induced by these is complex and depends on the structure of the downstream optimization problem (Elmachtoub and Grigas, 2022). Accordingly, different metrics that account for different uncertainty and accuracy evaluations can be useful, depending on the respective problem. We assess the used error metrics in RFD forecasting in Section 3.

The deterministic formulations ignore forecast uncertainty and thus can lead to potentially bad solutions, if high forecast errors occur. The stochastic formulations can provide optimal solutions, given a correct uncertainty estimate and are accordingly more robust to forecast errors. There are several approaches for optimization under uncertainty like robust optimization, scenario optimization, and distributionally robust optimization (Ning and You, 2019). For robust optimization, we require forecasts of

Table 2

Summary of forecast requirements based on the described optimization problems.

Type of constraint	Probabilistic adaptation	Required forecast description	Related metrics
Soft constraint	Recourse formulation, stochastic programming in general	Distributions or samples thereof	CRPS, MIS, RMSE, ME, etc.
Hard constraint	Chance constraints	Quantiles	PIN

Table 3

Number of papers which provide forecasts that serve optimization in a predict then optimize setting per target variable.

Optimization Paradigm	Deterministic	Stochastic programming	Chance constraints	Scenario or robust Optimization	None
Evaluation	Point mean forecast	Predictive distribution	Quantile	Scenario forecast	No or in sample
Enterprise	2	–	–	–	2
OD Matrix	3	2	–	–	2
Area	3	1	1	–	2
Terminals	1	–	–	–	–
Products	1	1	1	–	–
Others	2	–	–	–	–

Publications can appear in multiple cells, if they fulfill multiple criteria.

Blumenfeld and Beckmann (1988), Gao et al. (2010), Lin et al. (2018), Assemi et al. (2019) did not fall in any category.

valid scenarios, i.e., these should again reflect on the predictive distribution of the forecasted variable. Stochastic optimization involves a variety of formulations, some relying on samples of the relevant distributions, for example, in the sample average approximation (Kleywegt et al., 2002). In each case, appropriate (uncertainty) forecasts are central. However, these are themselves a forecasting task, including estimation errors (Chatfield, 2000). Hence, we argue that even for probabilistic models, theoretical optimality is not a sufficient evaluation scheme. Instead, decisions and their actual cost should be evaluated on empirical data, including some rolling origin evaluation scheme where forecasts are generated for different realizations of covariates along time (Tashman, 2000). This is necessary to account for the difficulty in generating accurate predictive distributions.

Table 3 links the optimization paradigms to the provided forecasts of the publications in this review. No publication evaluates scenario forecasts, and the connection remains weak in the other areas. The current RFD forecasting literature does not evaluate uncertainty estimation in forecasting. For instance, no publication uses the Mean Interval Score or the Pinball loss for evaluation of specific quantiles of the predictive distribution (see, Section 3). More generally, a large number of forecasting studies connected to RFD overly focus on the point mean forecast (the conditional expectation), while the optimization that typically follows requires richer information of the predictive distribution (Athanasopoulos and Kourentzes, 2022).

Uncertainty estimates are addressed by seven forecast publications. Al Hajj Hassan et al. (2020) introduce a parameter in their forecast combination to assign different costs to over- and under-forecasting, however, this is not empirically evaluated. Similarly, Karis (2019) evaluates prediction intervals that are based on asymmetric cost for over- and under-forecasting and Ren et al. (2020) estimate the conditional critical fractile for a news vendor problem. Deutsch et al. (1994) show that picking the optimal quantile of the demand distribution improves the inventory routing solution problem over just taking the expected demand. Generally, this can be done with quantile regression, by using appropriate loss functions, such as the pinball, when estimating a model, or empirically (Saoud et al., 2022). Chen and Lin (2009) estimate the uncertainty of product-level demand forecasts with a generalized autoregressive conditional heteroskedasticity (GARCH) model. Miori (2009) estimates a static negative binomial demand distribution to serve a truckload routing problem, but does not evaluate it out of sample. Garrido and Mahmassani (2000) estimate a multinomial probit model that provides a demand distribution, dependent on space, time, and socio-economic variables.

3. Evaluation of forecast experiment settings

In operations, the value of forecasting is determined by its support for decision making. We can obtain evidence about the performance of a predict then optimize solution by ex-post simulation of the decisions, given historical data and forecasts. Three publications use this approach (Deutsch et al., 1994; Chen and Lin, 2009; Cleophas and Ehmke, 2014). These simulations can, however, be complex (or even impossible) to implement and computationally expensive (Kourentzes et al., 2020). Hence, often, forecast accuracy metrics are used as proxies, assuming that more accurate forecasts improve decisions. Even though widely adopted in research and practice, this approach comes with pitfalls and better forecast accuracy does not linearly translate to lower cost for decisions (Fildes and Kingsman, 2011; Kourentzes et al., 2020; Elmachtoub and Grigas, 2022). The discussion that follows in this section highlights issues in the existing literature and provides guidelines for future work in the intersection of RFD and forecasting.

When a metric is used for evaluation of a predict then optimize solution it should reflect upon the data characteristics and the decision problem at hand. For example, as shown in Section 2.2, the form of the forecast required by optimization varies and can be related to different metrics. Accordingly, we evaluate the experiment design along the dimensions: (i) metrics as proxy for decision making, (ii) appropriateness of metrics to data, (iii) sampling of errors, and (iv) intuitiveness of metrics. Next, we describe these dimensions in more detail and identify overarching weaknesses in the experiment design of the examined publications.

3.1. Metrics as proxy for decision making

There are bias metrics such as the Mean Error (ME), and accuracy metrics, like the Mean Squared Error (MSE). Both perspectives are important: Forecast bias can be more harmful to decisions than forecast variance and in sample unbiasedness does not guarantee the same for the out of sample (Chatfield, 2000; Sanders and Graman, 2016; Barrow and Kourentzes, 2016). None of the examined publications evaluates forecast bias. Accuracy metrics summarize the magnitude of forecast errors, in contrast to bias that measures the on average over- or underforecasting. Different metrics relate to different parts of the predictive distribution where, e.g., the MSE minimizes upon the mean of the predictive distribution and the Mean Absolute Error (MAE) upon its median (Gneiting, 2011a). Three publications use MSE, or its root (Garrido and Mahmassani, 1998; Mrówczyńska, 2013; Liu and Lu, 2008), Al Hajj Hassan et al. (2020) use MAE, and five publications evaluate both (Deutsch et al., 1994; Yuan et al., 2018; van der Meulen, 2019; Ren et al., 2020; Najafabadi, 2022). For the comparison of forecast methods, the metric should correspond to the desired part of the predictive distribution (Kolassa, 2020), making MSE more relevant to the problems described in Section 2, as these require the expectation of demand if they assume a point forecast.

As we show in Section 2, we often are not interested only in the mean or median of the predictive distribution, but want to evaluate the accuracy of forecast distributions. To evaluate predictive distributions, the Continuous Ranked Probability Score (CRPS) can be used. Further, to evaluate predictive intervals, the Mean Interval Score (MIS) is suitable, and for specific quantiles, the Pinball Loss (Gneiting and Raftery, 2007; Laio and Tamea, 2007; Gneiting, 2011b, see the online supplement). Karis (2019) and Ren et al. (2020) evaluate forecast intervals with coverage probability, the latter additionally evaluate mean prediction interval length. Garrido and Mahmassani (1998) evaluate, whether the predicted distribution differs significantly from a normal distribution with known parameters. Coverage probability can be misleading, because it does not consider the width of the distribution. Very wide prediction intervals would lead to 100% coverage, though such a forecast would not serve decision making. Otherwise, only point forecasts are evaluated.

For a summary and guidelines of error metrics to choose, we refer to Svetunkov (2022, chapter 2), Koutsandreas et al. (2022), and Athanasopoulos and Kourentzes (2022). Also see Elmachtoub and Grigas (2022) for the relation between forecast errors and actual cost in optimization.

3.2. Appropriateness of metrics for data

When modeling disaggregate data, intermittency, i.e., series with several periods of zero activity, are common. To evaluate intermittent time series, the MAE is not suitable as the median is zero in this case, and a forecast of zero is of no use for decision making (Teunter and Duncan, 2009). Cumulative error metrics offer a potential remedy (Kourentzes and Athanasopoulos, 2021; Athanasopoulos and Kourentzes, 2022), given that we are often interested in demand over some lead time, as for example in vehicle routing (Gendreau et al., 2014; Saoud et al., 2022). None of the publications consider intermittency explicitly in their choice of metric, though present, for example, in Garrido and Mahmassani (1998, 2000). Some optimization models (e.g., the stochastic dynamic vehicle allocation problem) require a forecast on a very disaggregate level, where data is often intermittent. If the forecast accuracy is evaluated across multiple time series with different scales, we need scaled or percentage error metrics to make forecast errors comparable. The former are preferable, as percentage errors are asymmetric, e.g., the widely used Mean Absolute Percentage Error (MAPE) penalizes overforecasting more than underforecasting, due to its percentage nature (Kolassa, 2020). Seven of the examined publications use MAPE or APE, four of them exclusively (Zhou et al., 2006; Zhou, 2017; Xu et al., 2018; Karis, 2019; Grzelak et al., 2019; Al Hajj Hassan et al., 2020; Kmiecik, 2021). For scaled metrics, one can use the Mean Absolute Scaled Error (MASE, Hyndman and Koehler, 2006), the Root Mean Squared scaled Error (RMSSE, Makridakis et al., 2022), or the scaled versions of the Mean Interval Score (MIS) or Pinball loss (Gneiting and Raftery, 2007). For a summary of the state of the art on predicting and evaluating predictive distributions, see Gneiting and Raftery (2007) and Kolassa (2023).

3.3. Sampling of errors

Independent of the choice of metric, errors should be calculated on a test set, since (hyper-) parameter estimation/optimization necessarily leads to overfitting on the training (and validation) set. Hence, reporting forecast accuracy on the training set provides very limited evidence on the forecasting performance. Methods providing the best in sample fit do not predict best out of sample in general (Tashman, 2000). Four publications evaluate in sample (Fite et al., 2002; Miori, 2009; Mrówczyńska, 2013; Mitkow and Świdorski, 2019) and for two publications the evaluation sample is unclear (Gao et al., 2010; Xu et al., 2018).

When considering time series, empirical data on the forecasting performance is generated alongside two dimensions: Evaluating on different time series with various characteristics and evaluating along time for a single time series. For strong evidence, we require both, as the former provides variability and generality across series and the latter across time, desensitizing to special events and specific phases of business (Tashman, 2000). Fourteen publications evaluate on less than ten time series, of those, seven examine

only a single time series (see Tables 2.1–2.5 in the online supplement). Seven publications use rolling origin evaluation to evaluate across time (Godfrey and Powell, 2000; Chen and Lin, 2009; Yuan et al., 2018; Grzelak et al., 2019; van der Meulen, 2019; Karis, 2019; Al Hajj Hassan et al., 2020). All of the publications work on past freight demand data, which might be censored if requests for transportation could not be met, for example, due to insufficient capacity. This can lead to underestimating the actual demand in the forecasts and in the evaluation. In out-of-capacity situations, the related observations might be treated as missing values (Sousa et al., 2025).

Although we recognize the challenges of sourcing real data, we need to be wary of the limitations.

The performance of newly proposed forecasting approaches needs to be evaluated in comparison to simple benchmarks, as those often times provide better forecast accuracy than elaborate models (Makridakis and Hibon, 2000; Armstrong, 2006). We recommend using at least the random walk model, exponential smoothing, and ARIMA (Autoregressive Integrated Moving Average). While the latter two are frequently used, the random walk model is not considered at all, even though it does not depend on any modeling or implementation choices and therefore offers an objective benchmark.

3.4. Intuitiveness

Depending on the purpose of the forecasting experiment, we can consider different metrics as intuitive. If we forecast for optimization, the size of the error as given by the Mean Squared Error (MSE) is of interest. The MSE is not on the same scale as the observations and hence difficult to interpret. When communicating forecast errors to business, percentage errors such as MAPE are intuitive, though they come with statistical drawbacks. Scaled errors can be non-intuitive, as they by design have no scale, but remain useful for model comparisons when multiple time series are relevant. Nonetheless, Athanasopoulos and Kourntzes (2022) argue that it is valid to interpret scaled errors as normalized. Relative metrics, e.g., to some benchmark, are also helpful for intuitive comparisons. Only Godfrey and Powell (2000) use this approach in RFD forecasting.

4. Findings on forecasting problems

In the forecasting literature RFD appears in various types, namely for: (i) the whole enterprise, (ii) the origin–destination-matrix, (iii) delivery areas, (iv) terminals, and (v) single products. We structure the literature along these types of RFD. For each type of RFD, we summarize the findings on data structure, predictive value of inputs and model choice. We identify mismatches on the interface to optimization, as they appear in the surveyed literature. Tables 2.1–2.5 in the online supplement give an overview of the reviewed papers per publication and type of RFD. Similar to the section above, our objective is to summarize findings and identify weaknesses in the forecast modeling and research methods, that can limit our confidence in the findings, and highlight recent work in the forecasting literature that can help overcome these.

4.1. Forecasting on enterprise level

At the highest aggregational level, RFD can be forecasted for a whole motor carrier. Mitkow and Świdorski (2019) consider driven kilometers. For Zhou et al. (2006), Zhou (2017), and Lu et al. (2007) the unit of freight demand are unclear. Fite et al. (2002) break the freight demand at the enterprise level further down to industrial and regional sectors, indicating a freight demand hierarchy (we expand on the use of hierarchies in Section 5). We summarize the key findings:

Data: The freight demand at the enterprise level exhibits monthly seasonality (Mitkow and Świdorski, 2019).

Inputs: To model the seasonality, dummy-variables for each month (Mitkow and Świdorski, 2019) and economic indices have been shown to be useful for forecasting freight demand Fite et al. (2002). Which of those economic indices are specifically relevant varies based on the specific industry (Fite et al., 2002). In that study 36 observations of monthly freight demand were used to assess the predictive power of 107 regressors (including lags and leads of those). The authors use stepwise linear regression in combination with a correlation threshold. This is problematic, primarily as it can select spurious features, due to the large number of regressors, small sample size, and bivariate nature of correlation analysis. The use of shrinkage estimators, such as lasso, is recommended in this case (Sagaert et al., 2018).

Common seasonalities between regressors and the target can lead to spurious correlation, if modeling is done on the “raw” data, as it seems to be the case in Fite et al. (2002). Including spurious features leads to misspecified models and accordingly wrong uncertainty estimates and poor generalization performance (Chatfield, 2000; Smith, 2018). Instead, it is good practice to first capture this information in a univariate function or with features, such as seasonal dummies, to guard against spurious connections (Sagaert et al., 2018). This becomes critical when we are interested in the predictive distribution.

Models: Multi Layer Perceptrons (MLP) performed better than exponential smoothing (ETS) and ARIMA on a test set of six observations. Linear forecast combination with optimal weights on ordinary least squares of these improved further (Zhou et al., 2006; Zhou, 2017). We cannot draw strong conclusions from that, since we lack a rolling origin evaluation, only a single time series was examined, and MAPE was used (see Section 3). For an overview on forecast combination and recent advances, see Wang et al. (2023). In general, Machine Learning models such as MLPs have no model assumption about the data generating process, and thus uncertainty estimates must be sourced empirically. However, it is possible to forecast the parameters of a distribution with a neural network (Salinas et al., 2020).

4.2. Forecasting the origin–destination matrix

The demand for freight can be described in an OD-Matrix. We describe the main findings in the following.

Data: The data between terminals can be intermittent, while the data between single customers or cities is mostly intermittent (Garrido and Mahmassani, 1998; Miori, 2009). For intermittent data, metrics are a specifically poor proxy for the value of the forecast for decision making. Instead, the simulation of decisions is required to obtain evidence on the performance of forecasting methods (Kourentzes, 2013, 2014). Miori (2009) shows for the FTL case that a negative binomial distribution can describe the data better, since they found overdispersion when fitting a Poisson distribution. Assuming a distribution can also be seen as forecasting and should be evaluated as such. Al Hajj Hassan et al. (2020) cluster time series based on the frequency of moves in the respective lane using k-means. Afterwards, they pick forecasting methods per cluster.

Inputs: Several authors find that calendar features have predictive power to forecast RFD in an OD-Matrix (see Table 2 in the online appendix).

Models: Exponential smoothing with calendar effects performs slightly better than ARIMA with the former being less computationally intensive to fit (Godfrey and Powell, 2000). This is in line with Karis (2019) who found that a Holt–Winters adaptation (a form of exponential smoothing) with calendar effects outperformed the standard Holt–Winters formulation, as well as a simple MLP, the TBATS (Trigonometric Box–Cox ARMA Trend Seasonal) model, and simple exponential smoothing with respect to MAPE. Godfrey and Powell (2000) frame the model development as forecasting the OD-Matrix, but evaluate on the total demand originating from cities, as those time series exhibit less noise. Garrido and Mahmassani (1998, 2000) do the same (see Section 4.3), but while the exponential smoothing formulation by Godfrey and Powell (2000) can in general be used to forecast the OD matrix, this is not so easy for Garrido and Mahmassani (1998, 2000). They use spatial characteristics that are not easily adaptable to the OD-Matrix. Models that work well on smooth data do not necessarily perform well on intermittent data, so their evaluation provides limited evidence for forecasting the OD-Matrix (Kourentzes, 2014; Teunter and Duncan, 2009). Al Hajj Hassan et al. (2020) combine forecasts based on their observed errors in an ad hoc formulation. Base forecasts with higher forecast error are weighed less. The weights are adjusted in an exponential smoothing manner over time, as in Bates and Granger (1969). In difference to Bates and Granger (1969), Al Hajj Hassan et al. (2020) consider more than two base forecasts. For intermittent data the forecasting literature proposes Croston's method and its variations (Croston, 1972; Syntetos and Boylan, 2001; Teunter et al., 2011), bootstrapping (Willemain et al., 2004), temporal aggregation (Kourentzes and Athanasopoulos, 2021), or novel intermittent demand models (Svetunkov and Boylan, 2023). Depending on the size of the dataset, global machine learning models may be a valid alternative (Salinas et al., 2020).

4.3. Forecasting areas

When dealing with delivery or loading requests, freight demand is characterized in the literature by the scale, the region in which it arises, and the time (Liu and Lu, 2008; Cleophas and Ehmke, 2014). Additionally, the order value might be an attribute (Cleophas and Ehmke, 2014).

Data: The data at single customers is very intermittent. Liu and Lu (2008) derive the demand for freight out of the demand for oil. They assume proportionality between the two. van der Meulen (2019) use linear regression to derive trucking hours out of the requested demand for delivery and pick up in this region. Garrido and Mahmassani (1998) assume a Poisson distribution for demand in counties. When the data is aggregated to RFD in a region (e.g., a county), it gets smoother and exhibits intradaily, daily, weekly, and annual seasonality, where the daily seasonality is the most pronounced (van der Meulen, 2019). In a predict then optimize setting, the required forecast granularity is given by the planning level. This, however, might be adjusted based on the forecastability of different levels (Athanasopoulos and Kourentzes, 2022).

Inputs: The first factor of a Principal Component Analysis (PCA) on socio-economic indices was significant to predict freight demand in counties (Garrido and Mahmassani, 2000). PCAs linearly project features to a subspace with lower dimensions, allowing for a compressed representation of those, while losing only little of the original information, and has been shown to be beneficial for forecasting (Stock and Watson, 2002). Including open orders of customers known at the time of forecasting (advanced demand information), and seasonal variables, improves forecast accuracy (van der Meulen, 2019). Known demand and uncertain demand may also be treated separately in optimization.

Models: To include the spatial characteristics of the problem in the model formulation (via Spatio Temporal Auto-Regressive (Integrated) Moving Average, STAR(I)MA) can lead to a more parsimonious model, while retaining the similar predictive power as ARIMA (Garrido and Mahmassani, 1998). STARIMA models are an extension of ARIMA models, which include auto-regressive and moving average terms of spatially related variables. For RFD forecasting these variables consist of freight demand that arises in the regions next to the one for which the forecast is generated. These are described in Deutsch et al. (1994) and Garrido and Mahmassani (1998). The aggregation to regions improves forecast accuracy (Garrido and Mahmassani, 1998; Liu and Lu, 2008). This suggests a hierarchical structure in the problem, that we explore in Section 5. In the thesis of van der Meulen (2019), a MLP outperformed other methods in five out of six time series on RMSE and three out of six time series on MAE. A Bayesian adjustment of forecasts, to include advanced demand information (orders of customers), improved the initial forecasts by 28% and 65%, on MAE for two time series. They further split the demand for one region according to the type of tank needed in nine time series. For those, a MLP outperforms the seasonally adjusted average. Then they use the forecast of the MLP to break down the regional forecast hierarchically. Ren et al. (2020) build a custom neural network that incorporates the spatio-temporal characteristics of freight volume, and report a substantial improvement over ARIMA and an extreme learning machine (a variant of Multi Layer Perceptron) with respect to MAE, RMSE, coverage and mean prediction interval length. They evaluate on only three time series and a single origin, which limits the insights we can draw from it.

4.4. Forecasting for terminals

We found two publications that forecast at the level of terminals. Grzelak et al. (2019) forecast the occupied pallet space in a terminal over time. The pallet space can be derived from shipments, which relates it to the RFD for Staff Scheduling in Cross-dock Terminals. Najafabadi (2022) forecast arriving trucks at a deep sea terminal at hourly granularity, to control various terminal operations.

Data: The occupied pallet space exhibits yearly and daily seasonalities (Grzelak et al., 2019). Calendar events are evident.

Inputs: Dummy variables for month and day of week have been found to increase the model fit (Grzelak et al., 2019).

Models: A Seq2Seq LSTM outperformed a simple LSTM and the sample average on a testset of four time series (Najafabadi, 2022). Grzelak et al. (2019) use a modification of MAPE with the forecast in the denominator, instead of the actual values, which complicates the interpretation (Green and Tashman, 2009).

4.5. Forecasting on product level

Freight demand arises usually out of the demand for some product that needs to be shipped to an end customer or to an inventory holding unit. Therefore, freight demand might be characterized as product demand and is linked to the general demand forecasting literature. This kind of problem formulation is only feasible if the transportation planning entity knows or controls the inventory at the distribution centers or point of sales, which is typically not the case for road freight transport enterprises. Alternatively, Assemi et al. (2019) forecast the days spent in the inventory of a central warehouse, until the examined products are consolidated for long-haul shipment.

Data: The structure of the data is highly dependent on the examined product. For retail demand forecasting, this has recently been examined by Theodorou et al. (2021).

Models: A STARIMA-forecast had higher accuracy than forecasting with a sample average (Deutsch et al., 1994). Assemi et al. (2019) reformulate the regression problem as a classification problem by discretizing the days spent in inventory into three categories and using the prediction of those as forecast. They find that the Random Forest has the best accuracy. The authors do not benchmark with a simple regression formulation. Chen and Lin (2009) use a GARCH model to estimate the variance of the demand time series to derive optimal inventories in the inventory routing problem.

4.6. Others

Six publications did not fall into any of the above groups. For Gao et al. (2010), Kmiecik (2021) and Yuan et al. (2018) the kind of forecasted freight demand could not be determined. Blumenfeld and Beckmann (1988) propose an analytical derivation of expected stops per load, given a demand distribution per customer in the vehicle routing problem. Tabib et al. (2022) forecast the expected truck utilization along the route and compare a Random Forest, Polynomial Regression, Support Vector Machine and a Multi Layer Perceptron. The former achieves the highest accuracy based on out of sample R-Square (can be understood as a scaled MSE). As inputs serve the location of the customer, the stop number in the journey, time of visit, and day of visit. If the routes are planned flexibly, as in Gendreau et al. (2014), this forecast formulation intertwines decisions made in the optimization problem and forecasting the uncertain demand. This leads to a more difficult forecasting problem in comparison to treating the two steps separately.

Bock (2017) links forecasting methods to planning stages and problems in logistics, based on a qualitative assessment of forecasting methods (without empirical error evaluation) and an interview study. Yuan et al. (2018) evaluate an extreme learning machine with various pre-processing steps. However, no benchmarks are provided.

5. A hierarchical framework for RFD

The optimization problems in road freight transport rely on RFD of different granularities, in terms of units of analysis, geographies, time, or forecast horizons. We refer to these, generally, as levels. The planning process dictates the required forecast level. So far, we considered individual RFD planning problems separately. However, for a firm to operate it must address several simultaneously. Considering plans independently simplifies modeling, but it implicitly enforces restrictions. As data is aggregated into different levels, parts of the information becomes attenuated, while other become more pronounced (aggregation acts as a moving average, Athanasopoulos et al., 2024). If exogenous information is considered, that information may be available only at some levels. In practice, various stakeholders may hold different information that is not shared across decision makers equally, adding further limitations in what is available for modeling (Kourentzes, 2022). These necessitate that forecasting models at different levels will differ in structure and information. This can result in forecasts that differ across levels, even without considering any forecasting model misspecification, which is to be expected. Ultimately, forecasts at different planning levels will be *incoherent*, i.e., if the disaggregate forecasts are aggregated, they will not agree with the aggregate ones. Therefore, the inputs to the various optimization models at different planning levels will differ, resulting in diverging plans that can introduce waste and additional costs. For example, a carrier might plan for a high-volume forecast for vehicle scheduling, but a medium-volume forecast for the related cross docks. That could result in an excess of planned vehicles and not enough staff to handle the incoming load.

We advocate the use of hierarchical forecasting that was developed to make forecasts coherent, and as such provide a common basis for subsequent planning decisions. Hierarchical forecasting can be understood as a post-processing step, that makes

independently generated forecasts coherent. These aligned forecasts become the inputs for the optimization step at each level. The reconciliation step that ensures forecast coherency, which is at the core of hierarchical forecasting, typically results in accuracy gains (Athanasopoulos et al., 2009; Hyndman et al., 2011; Athanasopoulos et al., 2017; Wickramasuriya et al., 2019; Kourentzes and Athanasopoulos, 2019; Panagiotelis et al., 2021), improving the quality of inputs to the optimization at each level, beyond any benefits that come from forecast coherency and the alignment of plans (Kourentzes, 2022). Finally, a reason for the success and popularity of hierarchical forecasting is its simplicity (Athanasopoulos et al., 2024). As a post-processing step, it uses the output of independent forecasting models at the various planning levels. This keeps model complexity low. The alternative would be to construct large multivariate models, spanning data of different granularities, that are hard to specify, estimate, and can be computationally demanding.

Carriers' different planning processes can be expressed as a cross-sectional hierarchy, where aggregation can happen across various demarcations such as geography, and a temporal hierarchy, where aggregation occurs across time and forecast horizons. Those can be integrated, forming a cross-temporal hierarchy (Kourentzes and Athanasopoulos, 2019). These different types of hierarchy are based on the same mathematical formulation.

Besides the theoretical benefits, there is ample empirical evidence of accuracy benefits of hierarchical forecasting in various domains. Hyndman et al. (2011), Wickramasuriya et al. (2019), and Panagiotelis et al. (2021) show this for tourism demand, which arises in a spatial hierarchy, Athanasopoulos et al. (2017) demonstrate it for the temporal hierarchy of accident & emergency calls, and Babai et al. (2022) review the literature on hierarchical supply chain forecasting and come to the same conclusion. (Athanasopoulos et al., 2024) provides a comprehensive review of the hierarchical forecasting literature, summarizing empirical evidence from various application areas, with accuracy benefits being a common finding. Kourentzes and Athanasopoulos (2019) evidences that temporal hierarchies, provide larger accuracy benefits compared to cross-sectional ones.

Despite these findings, and acknowledging the hierarchical nature of freight demand (i.e., Godfrey and Powell, 2000; Fite et al., 2002; Liu and Lu, 2008; Al Hajj Hassan et al., 2020), only van der Meulen (2019) links to the hierarchical forecasting literature that uses forecasts on multiple levels. However, they use a top down approach, where forecasts are generated at an aggregate level and then disaggregated. Instead of reconciling forecasts, this merely sets the preference to a single level, and lacks the expected accuracy benefits from hierarchical forecasting. Therefore, there is a disconnect between the practical needs in the industry and the RFD forecasting contributions.

5.1. RFD hierarchies

We provide some examples how required RFD forecasts can be mapped onto hierarchies. The most disaggregate representation of RFD in the LTL case are single shipments, which we describe by their origin and destination, as well as by the origin terminal, transfer terminal (if present) and destination terminal through which they are routed. Depending on the respective decisions, various aggregations of those shipments are relevant. RFD for the vehicle routing problem with stochastic demands is the sum of all shipments that need to be collected from the considered customer and brought to the considered terminal. The freight demand along one route is the sum of the demand customers along that route, which is required in the calculation of the vehicle routing problem (Gendreau et al., 2014, see Section 2.2).

RFD for staff scheduling in cross-dock terminals is the sum of all shipments that route through the considered terminal. RFD for the stochastic dynamic vehicle allocation problem is the sum of all shipments that flow between the considered terminals. As shipments adhere to a flow constraint, those forms of RFD are connected: The sum of RFD for the vehicle routing problem with stochastic demands equates the overall volume of collected, respectively delivered freight. Counting inbound and outbound traffic, half of the sum of freight volume that is handled in terminals equates the overall freight demand between terminals.

In Fig. 3 we give a visual representation of the hierarchy of RFD, as it is required by optimization, in the LTL case. The different levels of the hierarchy are on the vertical axis. The nodes in the network are on the horizontal axis. The white bars indicate which of the dimensions per row are relevant for the respective aggregation level. For example, in RFD for the vehicle routing problem with stochastic demands (pick up), we are interested in all shipments between an origin and an origin terminal. The gray bars mark the dimensions which are irrelevant for that level.

Carriers for full truck load do not use terminals. The planning problems and the RFD hierarchy becomes accordingly simpler, as only origin and destination are relevant. Based on their location a spatial hierarchy can be derived. For example, similar to Athanasopoulos et al. (2009), based on administrative groupings.

Planning levels for road freight transport enterprises exhibit a temporal hierarchy, where forecasts are generated on multiple temporal aggregation levels (e.g., day, week, month, quarter, etc.). These can then be aligned using temporal hierarchies. For example, the service network design problem needs a RFD forecast on relations between terminals over a whole operating season, the Stochastic dynamic vehicle allocation problem needs such a forecast some days ahead. Similarly in staff scheduling in cross-dock terminals we need to strategically train or acquire workers which requires a long term forecast, and as well we have weekly, daily and hourly scheduling for operations. Another use of temporal hierarchies is to model intermittency in time series. In Section 3 we identified several cases where intermittency in the time series can make forecasting more difficult for RFD, such as the OD-Matrix. By temporally aggregating, intermittent time series will inevitably become non-intermittent, greatly simplifying forecasting with gains in accuracy (Kourentzes and Athanasopoulos, 2021).

Similarly, the flexibility of hierarchies allows firms that have multimodal shipping to map the different transportation networks enabling aligned forecasting and planning across those. Kourentzes and Athanasopoulos (2019) propose using hierarchies to map cross-firm interactions, although this depends on information sharing agreements between firms. In that case, alignment of operational planning could happen between firms that operate different modes of a network.

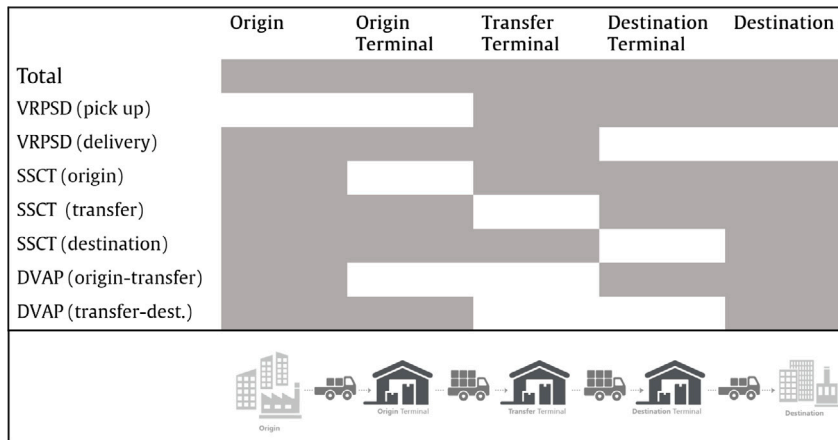


Fig. 3. The hierarchy of freight demand for a LTL-carrier. VRPSD: vehicle routing problem with stochastic demands, SSCT: staff scheduling for cross dock terminals, DVAP: dynamic vehicle allocation problem.

5.2. Modeling hierarchical forecasts

We introduce the prominent formulation of hierarchical forecasting to demonstrate its flexibility and highlight relevant questions for RFD forecasting. For this example, we consider freight demand for staff scheduling in cross-dock terminals and the overall freight demand. Both consist of sums of single shipments. We can link observations of freight demand on the bottom level with the upper levels of the hierarchy with a binary “summing” matrix S . The function of S is to map how the information from the bottom level $\mathbf{b}_t = (\text{shipment}_{AAGCE}, \text{shipment}_{BAGCF}, \dots)^T$ at time t is aggregated to form the observations at all levels $\mathbf{Y}_t$ so that

$$\mathbf{Y}_t = S\mathbf{b}_t \quad (1)$$

$$\begin{bmatrix} \text{overall} \\ \text{staff_scheduling_origin}_A \\ \text{staff_scheduling_origin}_B \\ \text{staff_scheduling_destination}_C \\ \text{staff_scheduling_destination}_A \\ \text{shipment}_{AAGCE} \\ \text{shipment}_{BAGCF} \\ \text{shipment}_{CBGAG} \\ \text{shipment}_{CBGAH} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \text{shipment}_{AAGCE} \\ \text{shipment}_{BAGCF} \\ \text{shipment}_{CBGAG} \\ \text{shipment}_{CBGAH} \end{bmatrix} \quad (2)$$

This holds for observations by construction. As discussed above, independent forecasts produced for the different series in $\mathbf{Y}_t$ will be incoherent. These are reconciled using:

$$\tilde{\mathbf{Y}}_t = S\mathbf{G}\hat{\mathbf{Y}}_t,$$

where $\tilde{\mathbf{Y}}_t$ are the hierarchically reconciled forecasts and $\hat{\mathbf{Y}}_t$ the initial independent ones. Matrix $\mathbf{G}$ has the role of combining all forecasts across the hierarchy to a new set of bottom level forecasts that account for the information in the hierarchy, which are then multiplied by S to be aggregated to the complete hierarchy, therefore supporting decisions at aggregate levels. The forecasts contained in $\tilde{\mathbf{Y}}_t$ are used as inputs to the various optimization models that support planning across levels, providing a common view of the future RFD.

The Matrix $\mathbf{G}$ is typically estimated as

$$\hat{\mathbf{G}} = (S^T \mathbf{W}^{-1} S) S^T \mathbf{W}^{-1},$$

where $\mathbf{W}$ is an approximation of the variance–covariance matrix of the unreconciled base forecast errors. In the literature, there are multiple approximations for $\mathbf{W}$ (Pritularga et al., 2021) and it can account for the level of the hierarchy (for scaling the forecasts) and the performance of the forecasts, with various simplifying assumptions to facilitate scalability. The reader is referred to Wickramasuriya et al. (2019) and Athanasopoulos et al. (2024) for detailed descriptions of the reconciliation step, and to Panagiotelis et al. (2022) for its probabilistic formulation. The forecasts in $\hat{\mathbf{Y}}_t$ can be sourced in any way, using statistics, machine learning, and expert domain knowledge.

This example has the objective to demonstrate the easiness of using hierarchical forecasting as a post-processing step of the independently generated forecasts, prior to the optimization for each planning level. The reconciled forecasts will be coherent and typically more accurate than the initial forecasts, and therefore, intuitively lead to superior plans. However, this intuition requires empirical validation. The implicit research question relates to how the different steps in the predict then optimize interact.

Forecasting models typically minimize a quadratic loss, between observations and forecasts. Forecast reconciliation minimizes incoherency, as the quadratic distance between forecasts at different levels. Depending on the planning level, the different optimization models will have a variety of objective functions. Each step minimizes a different loss. This weakens the aforementioned intuition, and also invites research for reconciliation strategies that address the RFD challenges, potentially matching the loss closer to the optimization step. This is not without its challenges, as the different levels in the hierarchy will have different optimization setups, constraints, and objective functions. The misalignment of losses exists even without the hierarchical consideration, however, limited to a single level. Likewise, when the different plans are optimized independently, the current formulations have no way to distinguish between incoherent and coherent forecasts, and therefore may harm the alignment of plans even if the forecasts contain the appropriate information. The realities of RFD planning problems invite the application of hierarchical forecasting, but at the same time open multiple research directions.

6. Conclusion

Forecasting is a vital part of planning for road freight demand enterprises. We reviewed the current state in the RFD forecasting literature, and its connection with the common optimization problems in practice, as formulated in the optimization literature. Our objective had been to summarize the findings, identify limitations by focusing on the requirements stemming from the interplay of forecasting and optimization/decision-making in RFD, and highlight promising avenues forward. Moreover, where the literature has been uniformly lacking, we provide guidelines to help researchers by summarizing the recent developments in the forecasting field. Furthermore, we argue that the current developments in hierarchical forecasting offer promising routes for research for RFD forecasting, and demonstrate how these could be formulated.

We find three significant research gaps: First, the current forecasting literature does not evaluate RFD forecasts as they are used in optimization. Mismatches arise on the forecast horizon, description of the predictive distribution, unit, and granularity. We show this for the vehicle routing problem with stochastic demands, staff scheduling in cross-dock terminals, the stochastic dynamic vehicle allocation problem, the (stochastic) service network design, the request acceptance problem, and the truckload routing problem (Section 2.1). Moreover, most of the literature does not serve any of the paradigms of optimization under uncertainty, with very little work on predictive distributions and predictive scenarios (Section 2.2). To address this, we underline the need for appropriate evaluation metrics and design. Otherwise, improvements reported in forecasting studies might not translate into improved decisions, because the evaluated quantities are detached from what is used for planning afterwards. Likewise, weaknesses in the design of forecast evaluation can reduce the validity of findings.

Second, RFD can be expressed as a cross-sectional and temporal hierarchy. This hierarchy arises out of the flow constraints in the business process, the spatial dimension of RFD, and planning on various horizons. The levels of this hierarchy relate to optimization problems. The current RFD forecasting literature, however, does not leverage the state-of-the-art hierarchical forecasting formulation that allows for probabilistic, more accurate, and aligned forecasts, and hence better decision making. If the hierarchy is ignored, different entities in the enterprise may plan on different bases, and accordingly produce non-matching decisions that lead to subsequent costs and waste.

Third, the surveyed optimization models assume that the demand distribution is known (or even deterministic). From a forecasting standpoint this is strongly contradicted. Crucially, uncertainty does not stem only from the stochastics of a process, but also from data and modeling limitations. Neglecting this might lead to underestimating the forecast uncertainty, and accordingly, suboptimal decisions. Likewise, considering the hierarchical structure of the decisions to be optimized can result in better estimates of demand uncertainty, due to the forecast reconciliation. Otherwise, this is another source of misestimation of uncertainty.

We identify some clear recommendations from this review for future research in RFD forecasting. Any forecasting contribution must be considered in the context of its application. This guides the choice of units, time granularity, forecast horizon, loss functions, and evaluation metrics. To help researchers, when an associated empirical case does not exist, this review provides an overview of these for different RFD decisions, motivated by optimization problems. For instance, demonstrating gains in point mean forecasts when an optimization model expects predictive distributions, or scenarios, will be of limited value and validity.

Nonetheless, we advocate for integrated forecasting and optimization solutions. When feasible, the gains from novel forecasting models should be evaluated on the decision outcome, typically after appropriate optimization or simulation. Likewise, optimization models do not benefit by assuming known or simplistic demand distributions and uncertainties, and future optimization-focused research could rely more, at minimum, on established forecasting models to obtain these. Integrated models, where forecasting and optimization are treated jointly, remain challenging, but this is an area that future research is welcome. Limited evidence from other application areas highlights that counter-intuitive findings can become evident only when these are investigated jointly (e.g., Kourentzes et al., 2020).

There are multiple methodological considerations in conducting research in predictive modeling. These are connected with appropriate and sufficient sampling, the use of adequate benchmarks and statistically valid metrics that are fit for purpose. To help researchers, we provide an overview of the considerations in the current forecasting discourse in Section 3. RFD has various features that demand dedicated forecasting research, and following these best practices can ensure the validity of the findings and help towards generalization. A striking omission and call for future work in the area is the use of modern machine learning and artificial intelligence methods that have shown promising performance in applications with similar data considerations as some of the RFD problems (for example, in the recent M4 and M5 forecasting competitions, and Kaggle competitions, Makridakis et al., 2020, 2022; Bojer and Meldgaard, 2021). These can be easily implemented as benchmarks or modified to adapt to RFD forecasting specifics.

A recommendation for future research focus is the use of hierarchical forecasting in RFD. We showed that it provides an intuitive mapping of associated decisions in RFD problems, enabling researchers to leverage its benefits directly. Beyond any forecasting benefits, considering the hierarchical structure of the problem motivates solving more realistic cases, where in practice the various decisions must be aligned for a firm to operate smoothly.

This review highlights a series of limitations when the two steps in the predict then optimize framework are done in isolation in RFD modeling. We expect to see similar disconnects and hierarchical modeling opportunities in other application areas of operations research, including other modes of transport like rail or ship. For example, empty container repositioning in sea freight (e.g., Meng and Wang, 2011). We welcome a closer integration of forecasting and optimization more generally, starting from ensuring that forecasts meet the requirements set by optimization problems and supported decisions. Our focus on this review has been on the forecasting side, with the optimization primarily examined to identify the requirements set on forecasts. An in-depth analysis of the RFD optimization models, looking at the extent to which forecasting outputs are fully utilized would be welcomed. Finally, our review focused exclusively on forecasting supporting road transportation. Other modes of transport introduce diverse forecasting challenges, due to their operations, network design, and relevant decision horizons, and therefore are out of scope. Nonetheless, the good forecasting practices discussed here remain relevant for all modes of transport.

CRedit authorship contribution statement

Benedikt Sonnleitner: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Nikolaos Kourentzes:** Writing – review & editing, Validation, Methodology. **Claudia Ehrig:** Writing – review & editing, Data curation. **Alexander Pflaum:** Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.tre.2025.104378>.

Data availability

No data was used for the research described in the article.

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